Magnetic circuits

Practice questions · Day 03 · Mon 26 Oct · pack M3

5. Practice

From your course files

PQ 3.3 (practice questions). Calculate the magnetic reluctance of the following objects:

a. A piece of circular iron wire, 1 mm in diameter and 10 cm long with a relative permeability of 1000.
b. An iron core of mean path length 20 cm, and cross-sectional area 2 cm² and relative permeability of 5000.
c. An airgap of length 1 mm and cross-sectional area 10 cm².

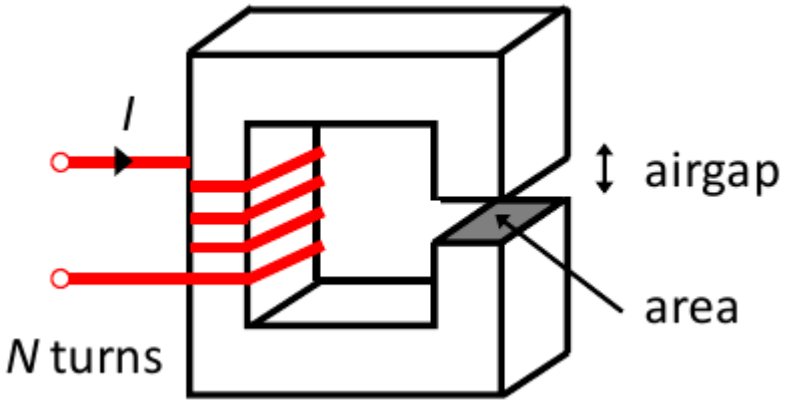
PQ 3.4 (practice questions). In the following cases, calculate the required mmf to produce a 1 T magnetic flux density in the object and hence calculate the required coil current if a coil of 100 turns is used to produce this mmf.

a. An airgap of length of 1 mm.
b. An magnetic core with a mean length of 20 cm and relative permeability 5000

Jun23 Q2a (past exam, 9 marks). *Electromagnetics.* Consider the iron c-core shown, which has a magnetic cross-sectional area of 20 cm² and a mean magnetic path length of 50 cm. The core has a winding with 32 turns and has a relative permeability of 3,000. The airgap length is 1.5 mm.

i. Calculate the reluctances of the airgap $\mathcal{R}_G$ and the core $\mathcal{R}_C$ and hence find the inductance of the coil. [4 marks]
ii. Calculate the maximum energy stored in the airgap only, if the maximum allowable peak core flux density is 1.8 T. [2 marks]
iii. The magnetic core is operating with a peak flux density of 1.5 T at 60 Hz. It has an iron loss of 12 W, of which 30% is hysteresis losses and 70% is eddy-current losses.

Calculate the new hysteresis, eddy-current and total iron loss if the peak flux density is reduced to 1.2 T, whilst the frequency is increased to 100 Hz. [3 marks]



Jun23 Q2a figure: iron C-core with an N-turn coil carrying I, an airgap, and the cross-section area marked (from the exam paper)

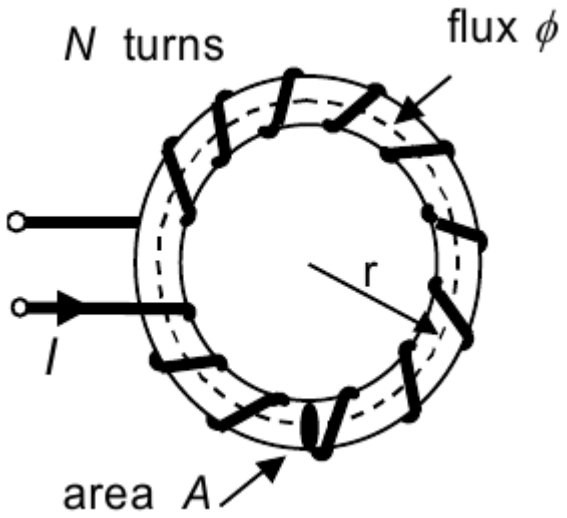
Jun24 Q2a (past exam, 7 marks). *Electromagnetics.* Consider a toroidal magnetic core which has a circular cross-section whose diameter is 2.5 cm. The core has a winding of 120 turns and can be assumed to have an infinite permeability. A fine saw is used to cut a thin slot to produce an airgap of 0.5 mm in one side of the core.

i. Calculate the reluctance of the airgap $\mathcal{R}_g$ and hence the inductance of the winding. [3 marks]
ii. If the maximum allowable peak core flux density is 1.5 T, calculate the maximum 50 Hz voltage which can be applied to the winding. [2 marks]
iii. Using a flux density of 1.5 T, calculate the force between the sides of the slot in the core. [2 marks]

The exam paper has no figure for this part. The sketch in the solution is drawn from the wording.

2023 quiz Q3 (6 marks). *AC Magnetic Circuits.* A winding is placed around a toroid iron core as shown in the figure on the right. The core has a mean diameter of 80 mm, a cross-section area of 20 cm² and a relative permeability μ_R of 1,000. The desired peak operating flux density of the core is 1.5 T.

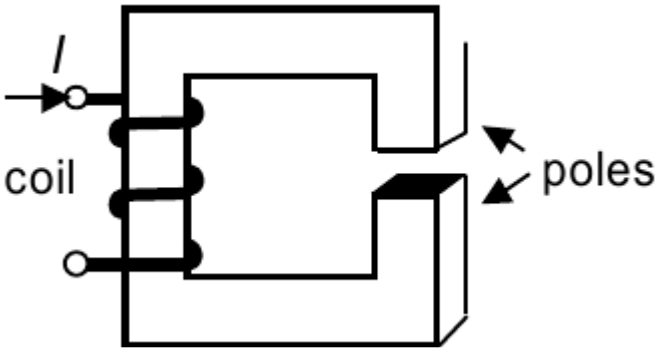
a. Given the winding has a rated voltage of 240 V at 50 Hz, use the back-emf equation to calculate the required number of turns. [2 marks]
b. For a different rated voltage, the number of turns is changed to 200. Calculate the winding inductance and reactance for this number of turns. [3 marks]
c. Based on the results from part (b), calculate the winding current magnitude when 100 V is applied to the winding. [1 mark]



2023 quiz Q3 figure: toroid with N turns carrying I, flux φ, radius r and cross-section area A (from the quiz paper)

2023 quiz Q4 (4 marks). *DC Magnetic Circuits.* An electromagnetic actuator has two iron poles with a 4 mm airgap between them, as shown in the figure on the right. Each pole has an area of 20 cm². A coil of 75 turns is used to create a flux density of 1.1 T in the airgap. Assume the core has infinite permeability.

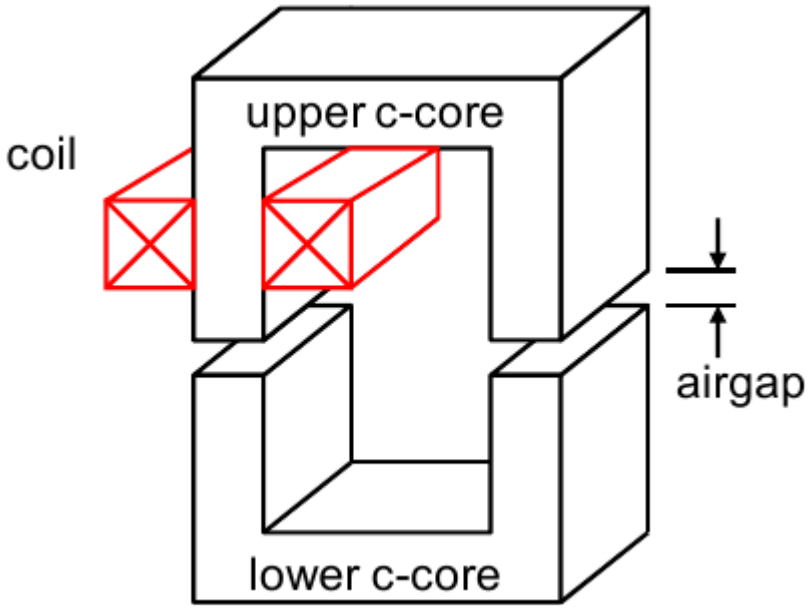
a. Calculate the required DC coil current. [2 marks]
b. Calculate the force of attraction between the poles. [2 marks]



2023 quiz Q4 figure: iron core with a coil carrying I and two poles across one airgap (from the quiz paper)

2024 test Q4 (7 marks). *Electromagnetics.* An electromagnet is constructed using two identical magnetic cores as shown on the right. The upper c-core is held stationary and magnetic attraction is used to hold the lower c-core. Each core has a mass of 0.3 kg, a magnetic cross-section area A of 5 cm² and a relative permeability of 4,000. The mean magnetic iron path in the cores for the flux is 25 cm (this is the total for the two cores). A coil of five turns carrying 3.5 A is placed on the upper core.

a. Assuming initially that the airgap between the cores is zero, what is the flux density in the magnetic circuit? [2 marks]
b. A piece of paper of 0.07 mm thickness is placed between the cores. The relative permeability of paper can assumed to be equal to 1. Calculate the new value of flux density in the magnetic circuit. [3 marks]
c. What is the minimum flux density in the magnetic circuit that would be required to support the weight of the lower core? Assume a value of acceleration due to gravity of 9.81 m/s². [2 marks]



2024 test Q4 figure: upper and lower C-cores with the coil on the upper core and the airgap between them (from the test paper)

2024 test Q5 (5 marks). *Magnetic Circuits.* A single-phase 240 V/48 V, 50 Hz transformer uses a laminated rectangular core with a cross-sectional area of 40 cm², magnetic path length of 0.8 m and relative permeability of 2,000. The peak operating flux density of the core should not exceed 1.6 T.

a. Using the back-emf equation, calculate the required number of primary turns for the transformer. [2 marks]
b. The primary winding consists of copper wire with a length of 40 m and a circular wire of diameter 0.5 mm. Find its resistance using a conductivity of $59 \times 10^6\ (\Omega\mathrm{m})^{-1}$. [2 marks]
c. Choose an appropriate number of turns for the secondary winding. [1 mark]

2023 quiz Q7 (4 marks). *Short-Answer Questions.* Answer the following question using words only (that is, without using sketches or equations).

In designing a transformer using a given magnetic core, what is the process for selecting the number of turns on the primary winding for a given primary voltage? Carefully explain the results of using too few turns. How about if too many turns are used?

PQ 3.10 (practice questions). Ferrite is a magnetic material which has a maximum allowable flux density before saturation of about 0.3 T. A circular ferrite core has a cross-sectional area of 5 cm² and has a small airgap cut in the core. A winding of 50 turns is placed on the core. Assume the ferrite has infinite permeability.

a. What length airgap is required to produce an inductance of 1 mH?
b. Alternatively, if the airgap is 1 mm, how many turns would be required to produce the same inductance? (number of turns needs to be an integer, hence round result to nearest)
c. For case (a), calculate the maximum current the inductor can carry without saturating.

PQ 3.13 (practice questions). A laminated magnetic core is operating at 50 Hz with a peak flux density of 1 T. At this flux density the core has an iron loss of 3 W.

a. The peak flux density is increased to 1.5 T. Estimate the new core loss.
b. The operating frequency is increased to 400 Hz with the original peak flux density of 1 T. Estimate the new iron loss assuming that (i) the core only has hysteresis loss and has no eddy-current loss, that is, the original 3 W of loss was only due to hysteresis. Hint: hysteresis loss is directly proportional to frequency. (ii) the core only has eddy-current loss and has no hysteresis loss (iii) the core has both eddy-current and hysteresis loss and at 50 Hz the eddy-current loss is equal to the hysteresis loss
c. In the situation described in part (b)iii, what is the required peak flux density at 400 Hz to give the same iron loss as operation at 50 Hz with a peak density of 1 T. Hint: remember that iron loss is proportional to B^2

PQ 3.14 (practice questions). Answer the following questions about magnetics.

a. Consider a magnetic core with an airgap. Where is the majority of energy stored?
b. Iron losses are caused by changing magnetic fields in a conductive magnetic material. Are there any iron losses when air is exposed to a changing magnetic field?
c. What are the two types of iron loss?
d. Which type of iron loss is more important when magnetic cores are operated using high frequency currents and why?
e. Why do motors and transformers normally use iron in the form of thin, insulated iron sheets rather than a solid core?
f. What happens to the input current of an inductor as the AC voltage is increased such that the core goes into saturation?

New questions

N1. Gapped rectangular core [5 marks]. A rectangular iron core has a mean path of 60 cm, a cross-section of 12 cm² and $\mu_R = 2500$. It has a single 0.8 mm air gap and a 250-turn coil.

a. Find $\mathcal{R}_C$, $\mathcal{R}_G$ and the coil inductance. [3]

b. Find the coil current for $B = 1.2$ T, and the fraction of the mmf dropped across the gap. [2]

Core for New question 2: shell core with a 200-turn coil and a 1 mm gap on the centre limb; the flux splits into the two outer paths

N2. Shell (E) core with parallel paths [7 marks]. A shell-type core has a 200-turn coil on its centre limb. The centre limb has length 10 cm and area 8 cm², and contains a 1 mm air gap. The flux returns through two identical outer paths. Each outer path has length 30 cm and area 4 cm². All the iron has $\mu_R = 2000$. Neglect fringing.

a. Draw the magnetic equivalent circuit and find the total reluctance seen by the coil. [3]
b. Find the coil inductance. [1]
c. Find the coil current needed for 1.2 T in the gap, and the flux density in each outer limb. [3]

N3. Back-emf design and loss scaling [8 marks]. A 230 V, 50 Hz iron-cored reactor has a core area of 15 cm², and its peak flux density must not exceed 1.5 T.

a. Find the number of turns and the actual peak flux density. [2]
b. At 230 V, 50 Hz the iron loss is 20 W, of which 40% is hysteresis. The reactor is moved to a 230 V, **60 Hz** supply. Find the new peak flux density and the new hysteresis, eddy-current and total iron loss. [3]
c. It is then used on a 40 Hz supply. Explain why 230 V is unsafe, and find the highest voltage that keeps $\hat{B} \leq 1.5$ T. Estimate the iron loss at that voltage. [3]

Lifter for New question 4: U-core with a 400-turn coil pulling an armature across two 1.5 mm gaps

N4. Electromagnetic lifter, two gaps [6 marks]. A U-shaped iron core with a 400-turn coil attracts an iron armature across two air gaps of 1.5 mm each. Each pole face is 6 cm². Treat the iron as infinitely permeable.

a. Find the coil current needed for a total pull of 150 N. [3]
b. Find the coil inductance and the energy stored at that current. [2]
c. The armature moves in until each gap is 0.5 mm, with the same current. Estimate the new force and comment. [1]

N5. Gapped inductor design, chained [8 marks]. You need a 5 mH inductor that carries a peak current of 10 A. The core has a cross-section of 6 cm², and its peak flux density must stay at or below 1.2 T. Assume infinite core permeability and a single air gap.

a. Using $\lambda = N\Phi = LI$, find the minimum number of turns (integer). [2]
b. Find the gap length that gives 5 mH with this number of turns. [2]
c. Find the actual peak flux density at 10 A, and show that $\frac{1}{2}L\hat{I}^2$ equals the gap energy. [2]
d. The inductor is used on a 50 Hz sinusoidal supply at a 10 A peak. Use the back-emf equation to find the rms voltage across it, and check it against $\omega L I_{\mathrm{rms}}$. [2]

6. Quick recall check

1. What are the units of reluctance, and what is the electric analogue of mmf?
2. Same coil and current, but the air gap (which dominates the reluctance) is doubled. What happens to L, to B, and to the force?
3. A transformer's applied voltage is halved at the same frequency. By what factor does the iron loss change?
4. Same voltage, but the supply changes from 50 Hz to 60 Hz. What happens to $\hat{B}$, P_H and P_E?
5. A winding is rated 240 V at 50 Hz. What is the highest voltage you should apply at 25 Hz, and why?